

Creating Immersive Future Living: An AI-Driven XR Interior Design System

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Figure 1: A user wearing a VR headset while exploring a interior environment representing with 3D Gaussian Splatting.

Abstract

We present an innovative immersive spatial design system that revolutionizes how users experience and create their ideal living

environments. Our system integrates cloud-based artificial intelligence, real-time ray-tracing rendering, and 3D Gaussian Splatting (3DGS) to deliver unprecedented "future living" experiences through XR/MR technologies. The system enables rapid design of ideal residential spaces, generates high-quality 3D scenes, and provides fully immersive exploration of future living scenarios, achieving the vision of "what you see is what you get" for future lifestyle experiences.

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CCS Concepts

• Computing methodologies → Rendering.

Keywords

Extended Reality, Interior Design, 3D Gaussian Splatting, Cloud AI, Real-time Rendering, Immersive Experience

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1 Introduction

1.1 Motivation and Vision

In an era of rapid technological advancement, people's imagination of ideal living spaces is often constrained by traditional design tools and visualization techniques. This project aims to break these limitations by leveraging advanced AI technology and immersive XR experiences, enabling users to truly "live" in their self-designed future homes, experiencing every detail and exploring every possibility.

1.2 Technical Contributions

Our system introduces three key innovations:

- **AI-Driven Spatial Design:** Cloud-based AI that understands user requirements and automatically generates personalized, structured spatial design solutions.
- **Photorealistic Real-time Rendering:** Cloud-based ray-tracing for real-time rendering with immediate preview of design modifications.
- **Seamless XR/MR Integration:** Conversion of spatial design data into high-quality 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] rendering for seamless XR/MR experiences.

2 System Architecture and Implementation

2.1 Core Experience Pipeline

Phase 1: Intelligent Requirement Capture. The system uses natural language processing and AI to analyze user lifestyles, habits, and needs from text, speech, or image inputs. A prompt library guides suggestions, while AI generates optimized, personalized design concepts with tailored layouts and styles.

Phase 2: Collaborative Design Optimization. Human-AI collaboration refines design details with real-time visualization. AI recommends furniture, decor, and color schemes, ensuring efficiency and coherence.

Phase 3: Immersive Living Experience. High-fidelity 3DGS-rendered virtual spaces allow free navigation and interaction. Simulation of real-world scenarios validates designs before implementation.

2.2 Technical Framework

Cloud-AI Processing Layer.

- Natural language understanding for design requirements.
- Generative AI models for spatial layout optimization.

Real-time Rendering Engine.

- Cloud-based ray-tracing for photorealistic visualization.
- Dynamic scene updating for design modifications.
- Optimized rendering pipeline for XR device streaming.

3DGS Integration Module.

- Conversion of design data to 3D Gaussian representations.
- High-fidelity scene reconstruction for immersive experiences.

3 Evaluation and Impact

3.1 XR Integration Benefits

XR integration enables users to freely experience and explore diverse spaces and lifestyles beyond physical constraints, blurring the boundaries between virtual and real environments. At the same time, it redefines interaction paradigms by moving beyond traditional interfaces toward natural, intuitive engagement with digital content, realizing the vision of "think it, see it, get it."

3.2 XR Vision Alignment

This project advances XR by integrating digital and physical environments to overcome spatial constraints, enabling immersive experiences at any scale. As virtual and real spaces merge, boundaries fade, forming a unified and adaptive spatial interface.

It replaces traditional inputs with intuitive interactions, merging digital and physical experiences to enhance immersion and collaboration. Future developments will enable users to create AI-driven spatial designs via voice commands entirely within the XR environment.

3.3 Societal Value and Impact

Design Democratization. This approach empowers individuals to design their own living spaces, lowering barriers to quality design while unlocking collective creativity.

Sustainability Advancement. Virtual design validation minimizes prototyping waste, while accurate previews reduce renovation rework. Moreover, it fosters environmentally rational consumption, aligning design with ecological sustainability.

4 Conclusion and Future Work

This project demonstrates the potential of combining cloud AI, real-time ray tracing, 3DGS, and XR/MR under the theme of "Creating Immersive Future Living," establishing not just technical advances but a new paradigm of lifestyle experience. It reflects XR's vision of enhancing daily life by integrating digital and real worlds, shifting users from passive viewers to active co-creators.

Future work will enhance AI personalization through extended interaction learning, support multi-user co-design, integrate IoT-based smart home simulations, and expand datasets to include diverse cultures, lifestyles, and housing types. These efforts will help create more adaptive and inclusive design environments.

References

- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.